# Name :Khansa javeed

# Roll num:242201048

# OPERATING SYSTEMS LAB

Lab Assignment #1 — BSCS-5A

## Task 1 — Windows & Keyboard Shortcuts

As a system administrator, keyboard shortcuts reduce dependence on the mouse and save time. I demonstrated 10 shortcuts; 7 of them use the Windows key.

### 1. Win + E

**Purpose:** Opens File Explorer instantly for quick file and folder navigation.

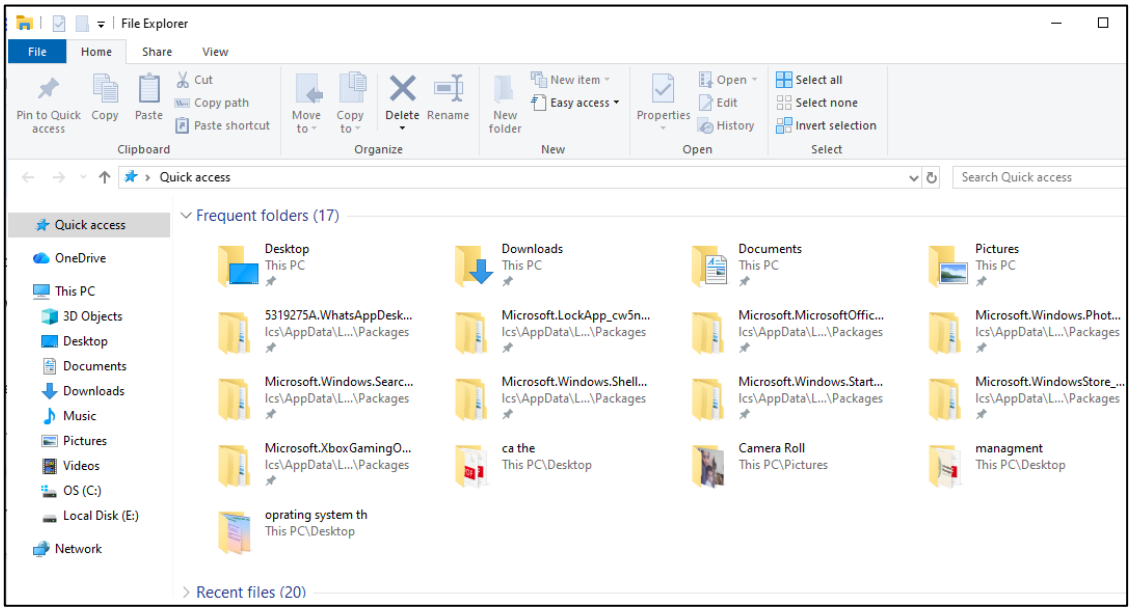



*Figure: File Explorer opened with Win + E*

## 2. Win + R

**Purpose:** Opens the Run dialog to quickly launch programs and system utilities such as cmd and powershell.

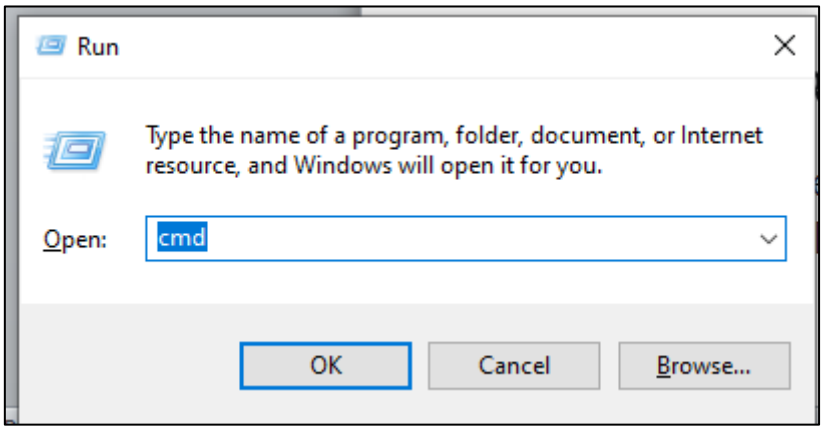



*Figure: Run dialog opened with Win + R*

## 3. Win + L

**Purpose:** Locks the workstation immediately to secure the system when stepping away.

## 4. Win + I

**Purpose:** Opens Windows Settings for quick access to system configuration.

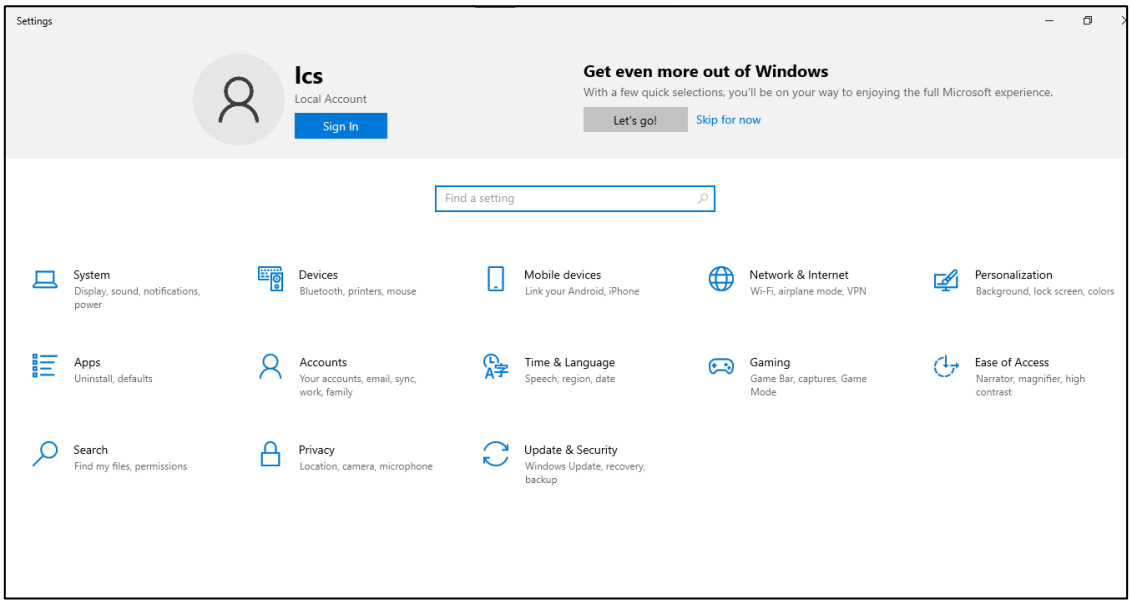



*Figure: Settings opened with Win + I*

## 5. Win + X

**Purpose:** Opens the Quick Link menu with administrative tools such as Device Manager, Disk Management and Terminal.

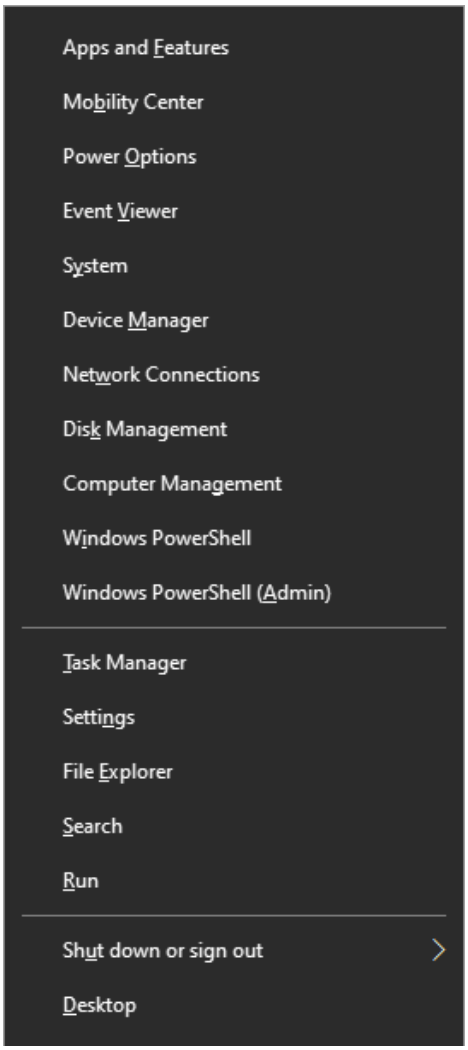



*Figure: Quick Link menu opened with Win + X*

## 6. Win + D

**Purpose:** Minimizes all open windows and shows the desktop.




*Figure: Desktop shown with Win + D*

## 7. Ctrl + Shift + Esc

**Purpose:** Opens Task Manager directly to monitor and manage running processes.

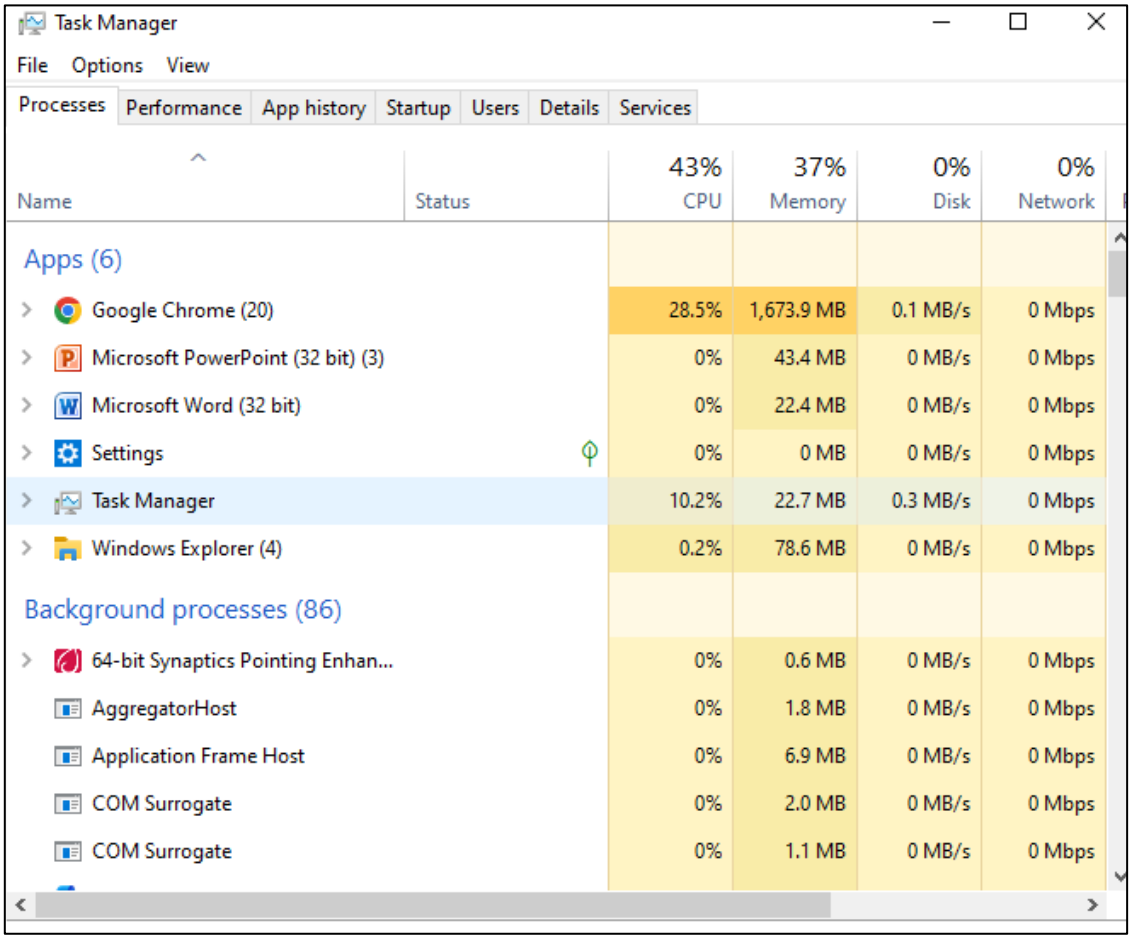



*Figure: Task Manager opened with Ctrl + Shift + Esc*

## 8. Alt + Tab

**Purpose:** Switches quickly between open applications.

## 9. Win + S

**Purpose:** Opens Windows Search to find apps, files and settings rapidly.

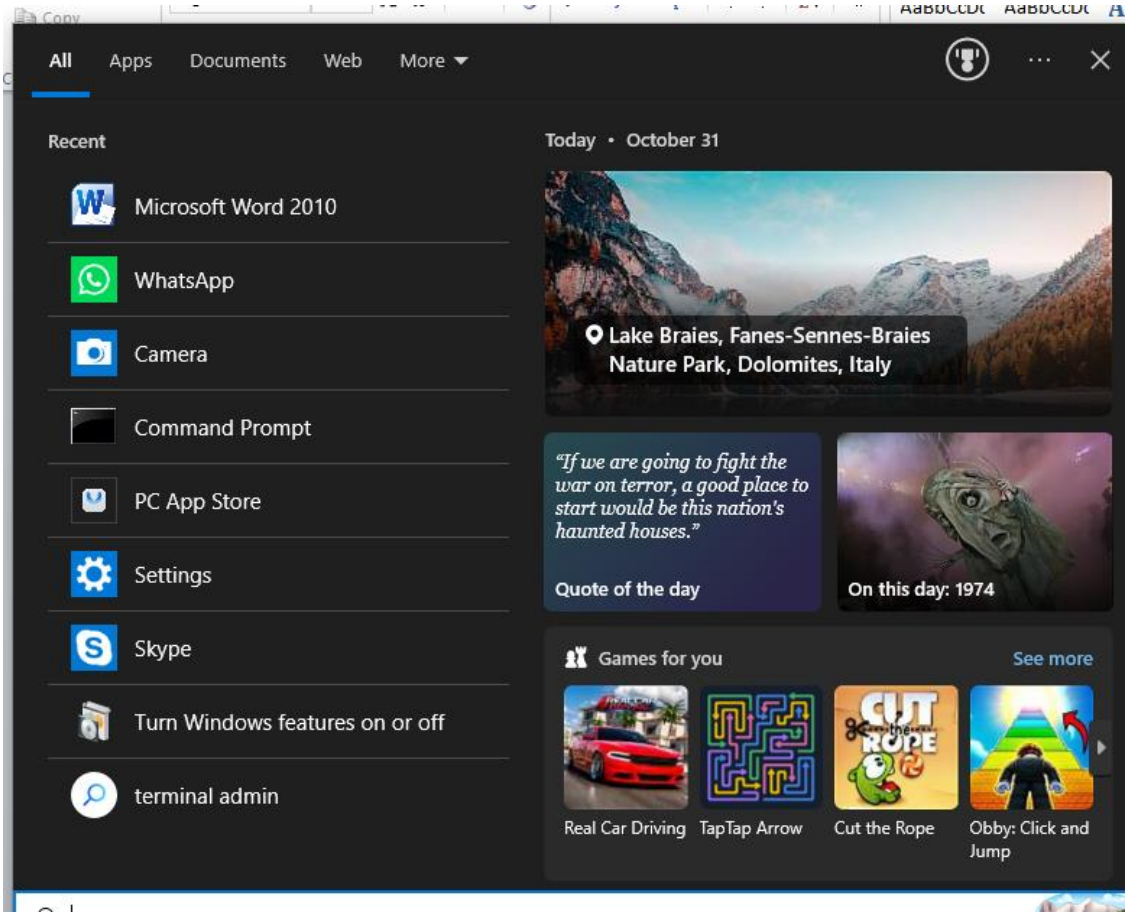



*Figure: Search opened with Win + S*

## 10. Win + PrtScn

**Purpose:** Captures the full screen and saves the screenshot automatically in the Pictures folder.

# 3. Task 2 — File & Directory Management Using CMD

Command Prompt was opened with Win + R → cmd. File Explorer was not used for any step.

## Step 1: Create the directory structure and display it

```
mkdir OS_Lab_Assignment
cd OS_Lab_Assignment
mkdir Students Reports Backup Scripts
mkdir Reports\Weekly
mkdir Reports\Final
tree
```

mkdir creates directories; tree displays the complete directory structure. The prompt shows the current directory C:\Users\lcs\OS_Lab_Assignment (the cd command without arguments also prints it).

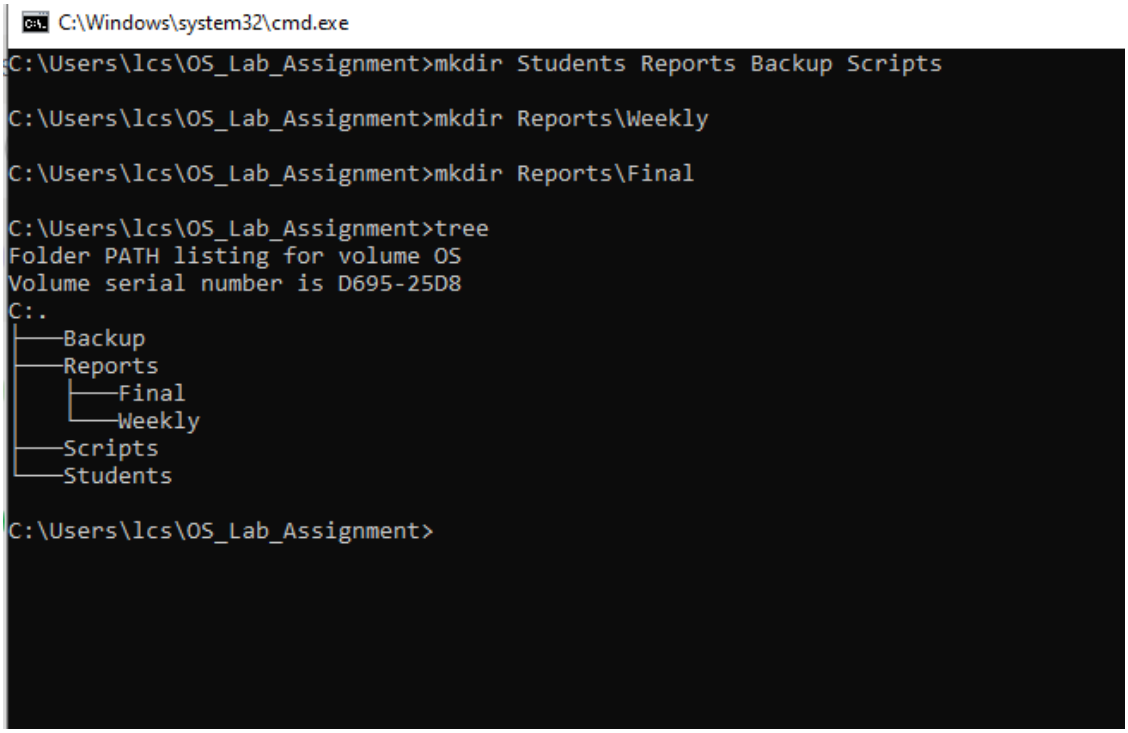



*Figure: Directory creation commands and tree output*

## Step 2: Create student_info.txt, write details and display it

```
echo Name: Khansa Javeed > student_info.txt
echo Registration Number: BSCS-48 >> student_info.txt
echo Semester: 5th >> student_info.txt
echo Section: B >> student_info.txt
type student_info.txt
```

The > operator creates the file with the first line and >> appends the next lines; type displays the content.

```
C:\Users\lcs\OS_Lab_Assignment>echo Name:khansa javeed > student_info.tx

C:\Users\lcs\OS_Lab_Assignment>echo Name: Ali Khan > student_info.tx

C:\Users\lcs\OS_Lab_Assignment>echo Semester: 5th >> student_info.txt

C:\Users\lcs\OS_Lab_Assignment>
C:\Users\lcs\OS_Lab_Assignment>echo Section: B >> student_info.txt

C:\Users\lcs\OS_Lab_Assignment>type student_info.txt
Semester: 5th
Section: B

C:\Users\lcs\OS_Lab_Assignment>
```

*Figure: Writing details into student_info.txt and displaying it with type*

## Step 3: Copy the file into Backup and rename it

```
copy student_info.txt Backup\student_info_backup.txt
dir
```

The file was copied into the Backup folder directly under the new name student_info_backup.txt (copy + rename in one command).

```
C:\Users\lcs\OS_Lab_Assignment>copy student_info.txt Backup\student_info_backup.txt
        1 file(s) copied.

C:\Users\lcs\OS_Lab_Assignment>dir
 Volume in drive C is OS
 Volume Serial Number is D695-25D8

 Directory of C:\Users\lcs\OS_Lab_Assignment

31-Oct-2026  11:28 pm    <DIR>          .
31-Oct-2026  11:28 pm    <DIR>          ..
31-Oct-2026  11:23 pm    <DIR>          Backup
31-Oct-2026  11:24 pm    <DIR>          Reports
31-Oct-2026  11:23 pm    <DIR>          Scripts
31-Oct-2026  11:23 pm    <DIR>          Students
31-Oct-2026  11:28 pm                17 student_info.tx
31-Oct-2026  11:28 pm                29 student_info.txt
               2 File(s)             46 bytes
               6 Dir(s)  62,450,814,976 bytes free
```

*Figure: copy command and directory listing*

## Step 4: Display all files and folders

```
dir
dir Backup
```

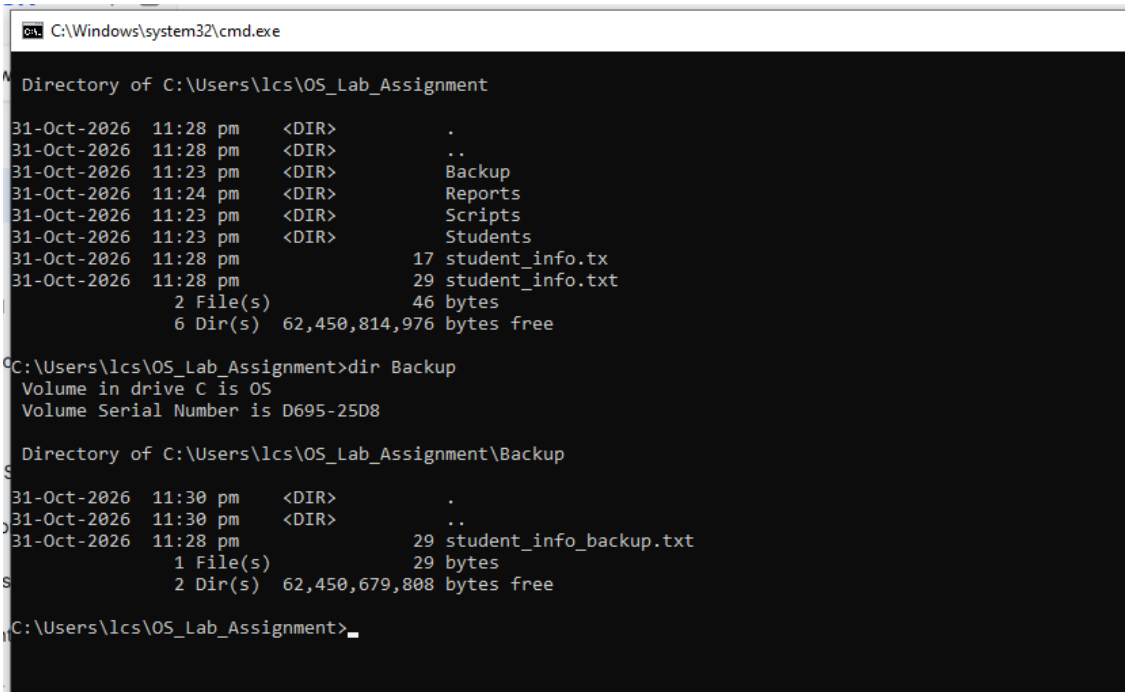



*Figure: dir of OS_Lab_Assignment and dir Backup showing student_info_backup.txt*

# 4. Task 3 — CMD System Investigation

Format for each item: Command → Purpose → Output/Screenshot → Short Explanation.

## 3.1 hostname

**Command:** hostname

**Purpose:** Find the computer name.

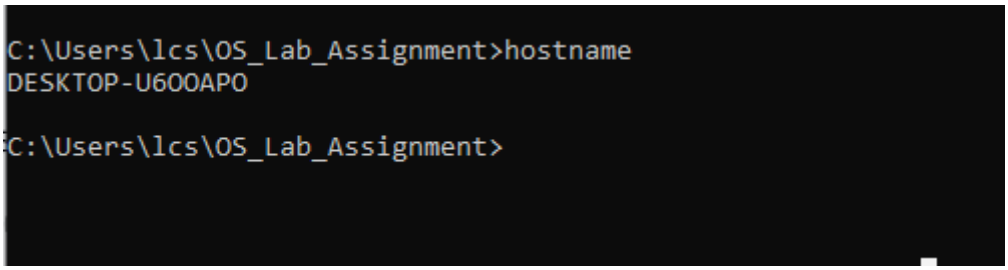



*Figure: Output of hostname*

**Explanation:** The computer name is DESKTOP-U6OOAPO.

## 3.2 whoami

**Command:** whoami

**Purpose:** Find the current username.

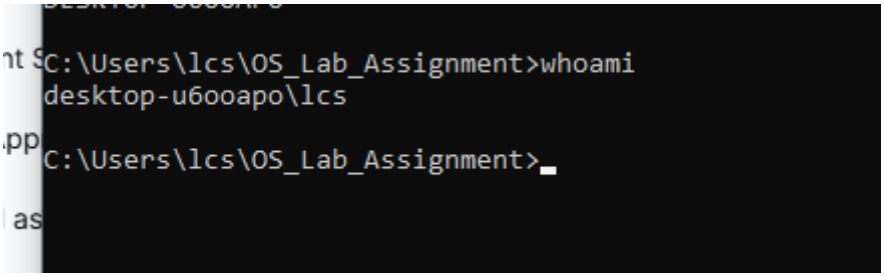



*Figure: Output of whoami*

**Explanation:** The logged-in user is lcs on machine desktop-u6ooapo (format: computer\user).

## 3.3 ipconfig

**Command:** ipconfig

**Purpose:** Show Windows IP configuration and the current IP address.

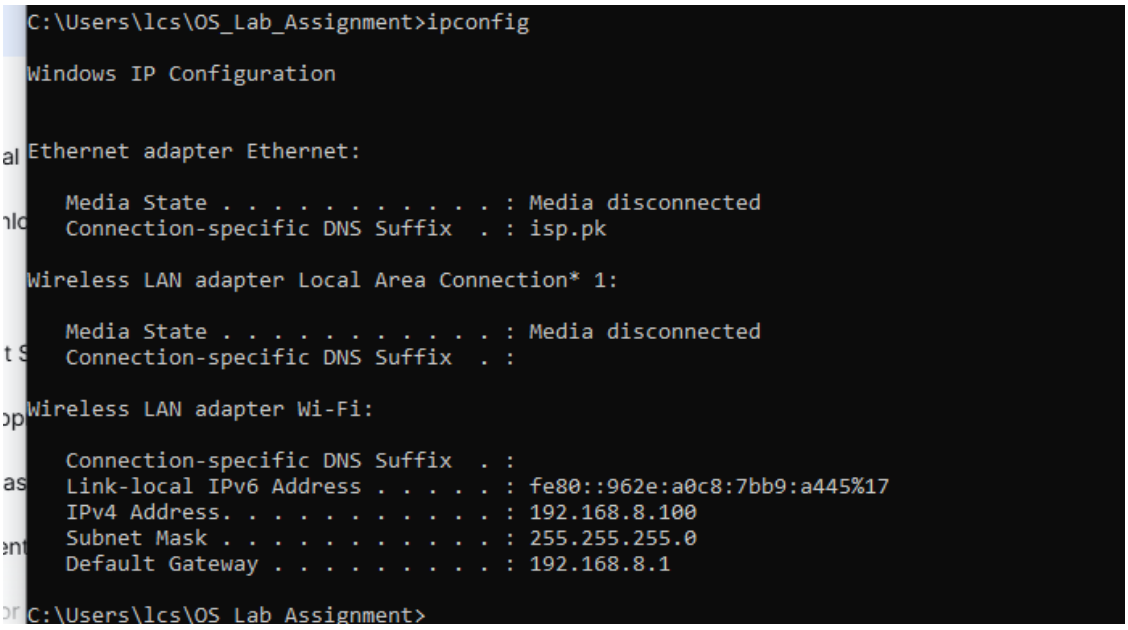



*Figure: Output of ipconfig*

**Explanation:** The active Wi-Fi adapter has IPv4 address 192.168.8.100, subnet mask 255.255.255.0 and default gateway 192.168.8.1. Ethernet and the other adapter show 'Media disconnected'.

## 3.4 ping google.com

**Command:** ping google.com

**Purpose:** Check network/internet connectivity.

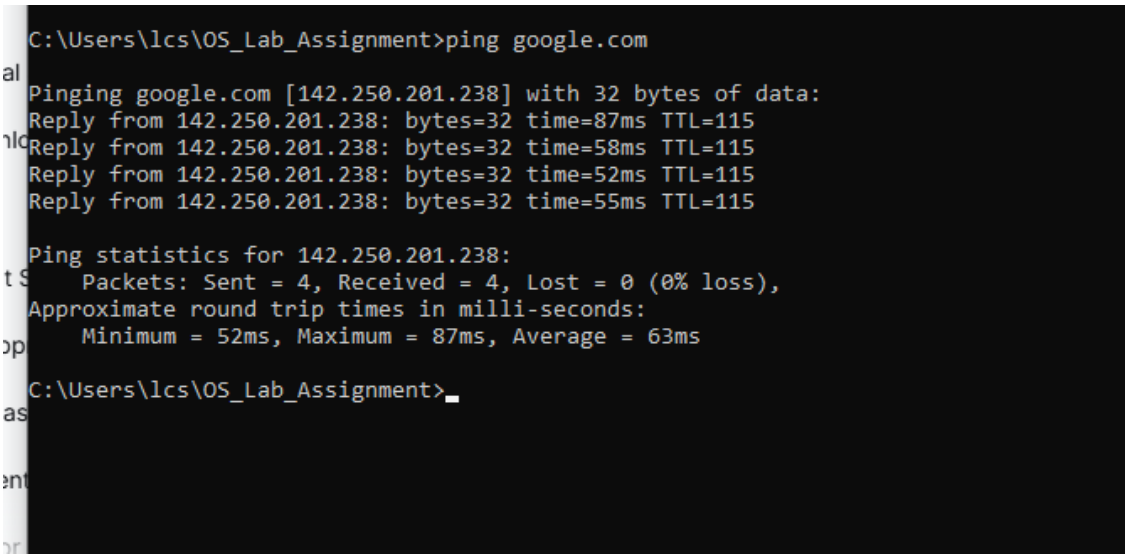



*Figure: Output of ping google.com*

**Explanation:** 4 packets were sent and 4 received with 0% loss (average 63 ms), so the computer is connected to the internet.

## 3.5 tasklist

**Command:** tasklist

**Purpose:** Display running processes.

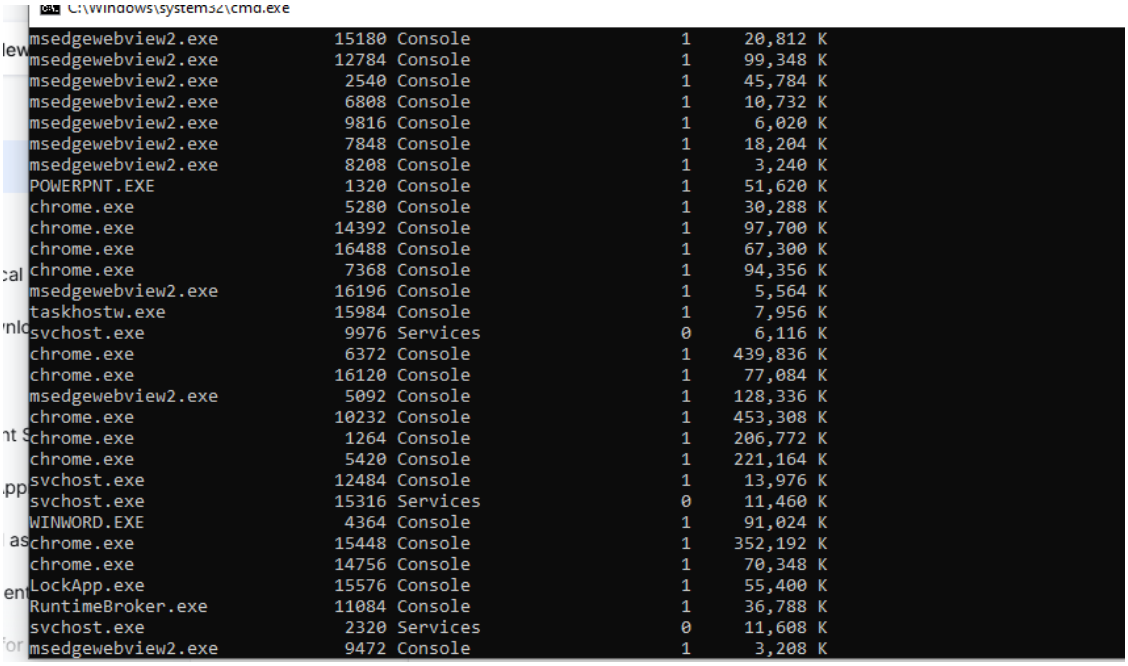



*Figure: Output of tasklist*

**Explanation:** Lists every running process with its image name, PID, session and memory usage (e.g. chrome.exe, svchost.exe, WINWORD.EXE).

## 3.6 date /t and time /t

**Command:** date /t and time /t

**Purpose:** Display the current date and time.

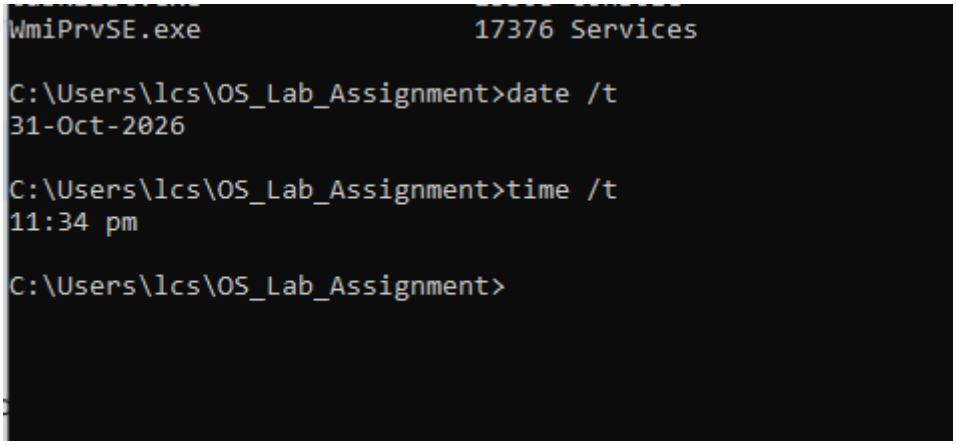



*Figure: Output of date /t and time /t*

**Explanation:** The system date is 31-Oct-2026 and the time is 11:34 pm.

## 5. Task 4 — Batch Script Automation

I created system_check.bat with Notepad (notepad system_check.bat) and ran it from CMD. The script is given below as code.

## Batch script: system_check.bat

```
@echo off
echo ==========================================
echo        WELCOME TO SYSTEM CHECK SCRIPT
echo   Name: Khansa Javeed   Reg No: BSCS-48
echo ==========================================
echo.
echo Current User  : %USERNAME%
echo Computer Name : %COMPUTERNAME%
echo Date          : %DATE%
echo Time          : %TIME%
echo.
echo ------- Network Information -------
ipconfig
echo.
if not exist "Daily_Backup" (
    mkdir "Daily_Backup"
    echo Daily_Backup folder created successfully.
) else (
    echo Daily_Backup folder already exists.
)
echo.
copy student_info.txt Daily_Backup\
echo student_info.txt copied to Daily_Backup successfully.
echo.
echo ==========================================
echo       SYSTEM CHECK COMPLETED SUCCESSFULLY
echo ==========================================
pause
```

## Execution of the script

```
system_check.bat
```

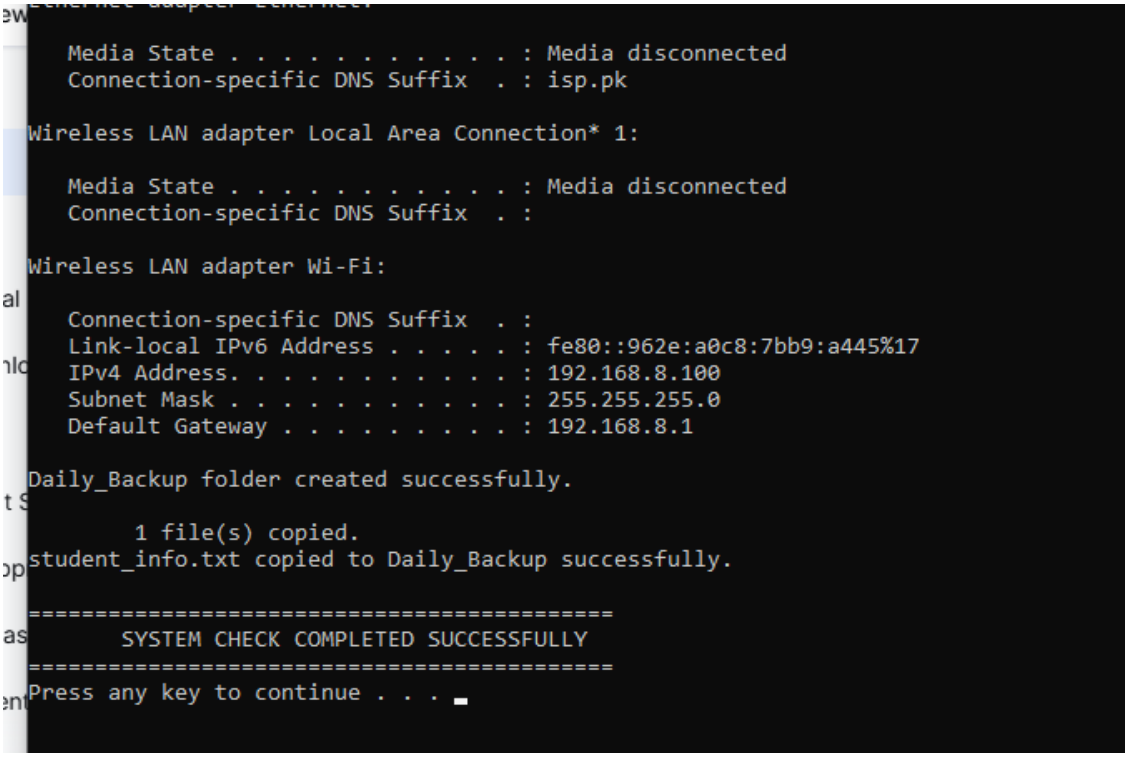



*Figure: Output of system_check.bat (network info, folder creation, file copy, completion message, pause)*

## Explanation of important commands

| Command | Explanation |
|---|---|
| @echo off | Hides the commands themselves so only their output is displayed. |
| echo | Prints text or variable values; echo. prints a blank line. |
| %USERNAME%, %COMPUTERNAME% | Environment variables holding the current user and computer name. |
| %DATE%, %TIME% | Built-in variables that give the current date and time. |
| ipconfig | Displays IP/network information. |
| if not exist ... mkdir | Creates the Daily_Backup folder only if it does not already exist. |
| copy | Copies student_info.txt into the Daily_Backup folder. |
| pause | Waits for a key press so the window does not close immediately. |

# 6. Task 5 — PowerShell Administration

PowerShell was opened from the Start menu / Win + X.

## 5.1 Display current working directory

```
Get-Location
```

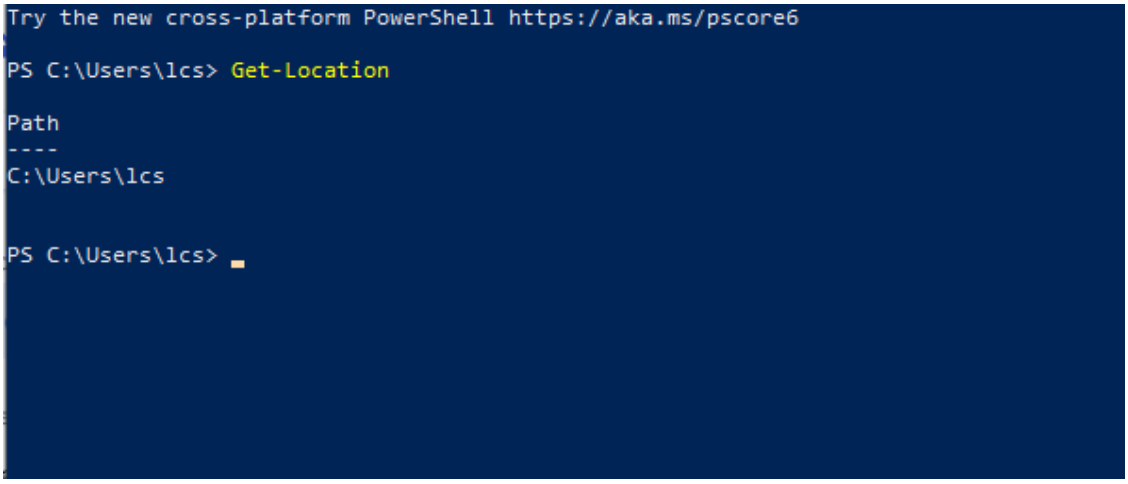



*Figure: Get-Location output*

**Explanation:** Get-Location shows the current path C:\Users\lcs.

## 5.2 List files and folders

```
Get-ChildItem
```

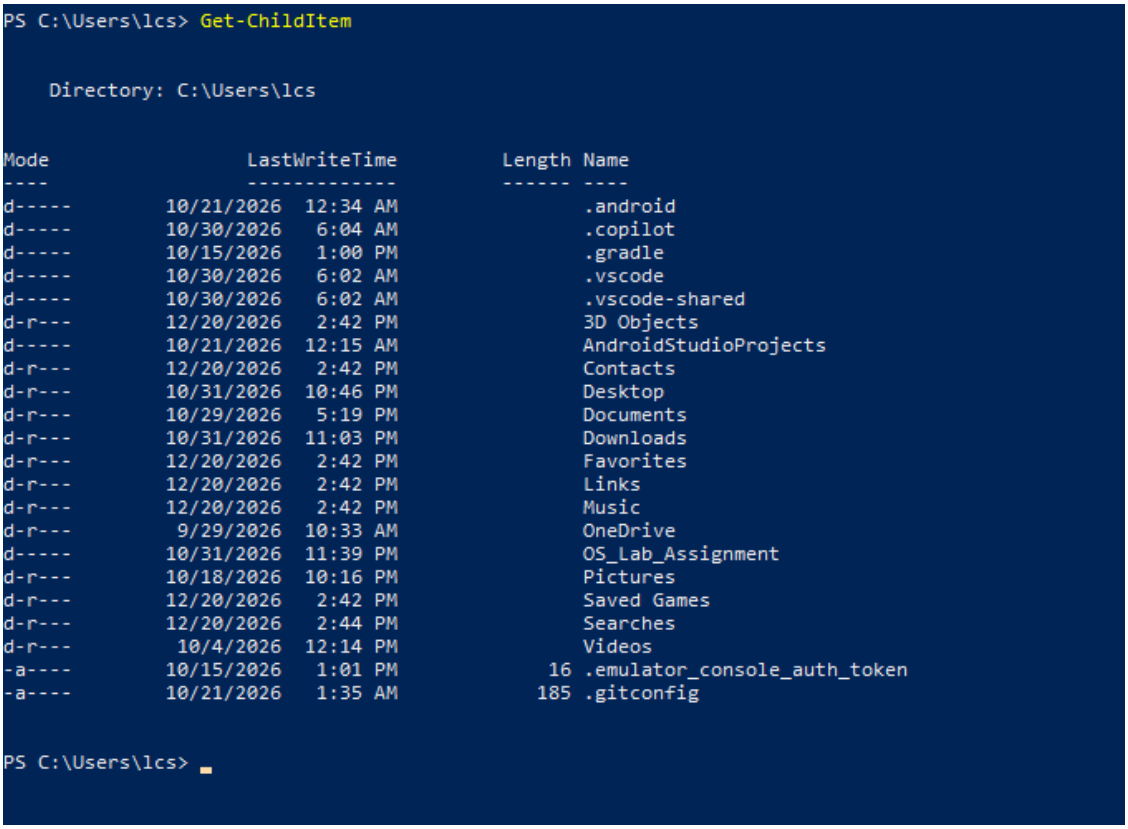



*Figure: Get-ChildItem output*

**Explanation:** Lists all files and folders (mode, last write time, length, name) of the current directory.

## 5.3 Create folder PowerShell_Task

```
New-Item -ItemType Directory -Name "PowerShell_Task"
```

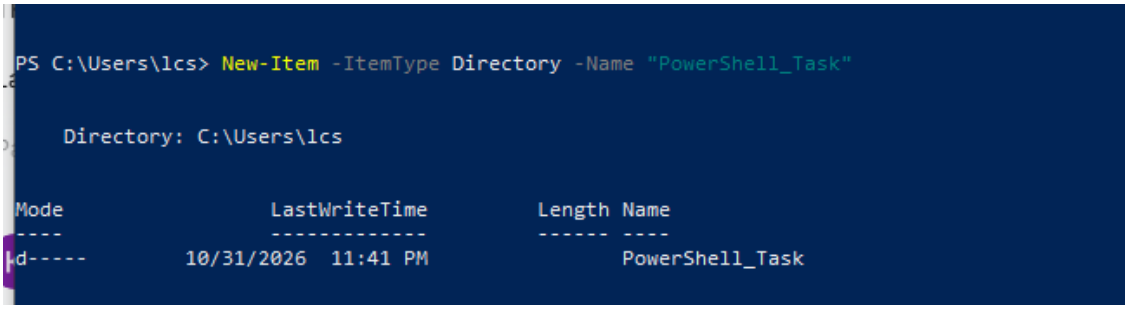



*Figure: Creating the PowerShell_Task folder*

**Explanation:** New-Item with -ItemType Directory creates a new folder.

## 5.4 Create a text file inside it

```
New-Item -ItemType File -Path "PowerShell_Task\info.txt"
```

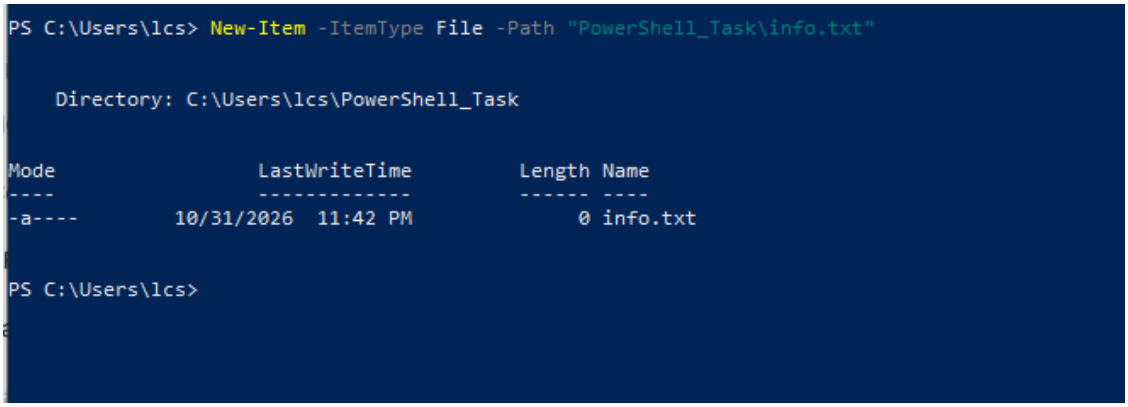



*Figure: Creating info.txt inside PowerShell_Task*

**Explanation:** Creates an empty text file named info.txt in the new folder.

## 5.5 Write Name & Registration Number and read the file

```
Set-Content -Path "PowerShell_Task\info.txt" -Value "Name: Khansa
Javeed`nRegistration Number: BSCS-48"
Get-Content -Path "PowerShell_Task\info.txt"
```

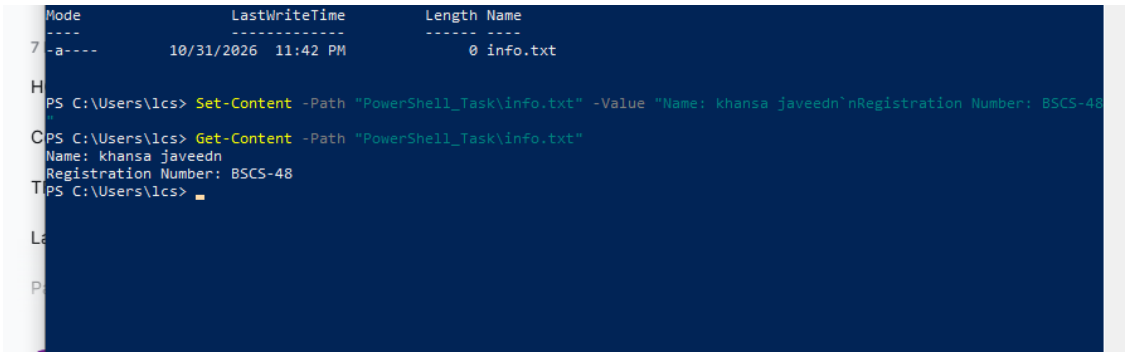



*Figure: Writing to info.txt and reading it with Get-Content*

**Explanation:** Set-Content writes the text into the file and Get-Content displays it, showing my name and registration number.

## 5.6 Display running processes

```
Get-Process
```

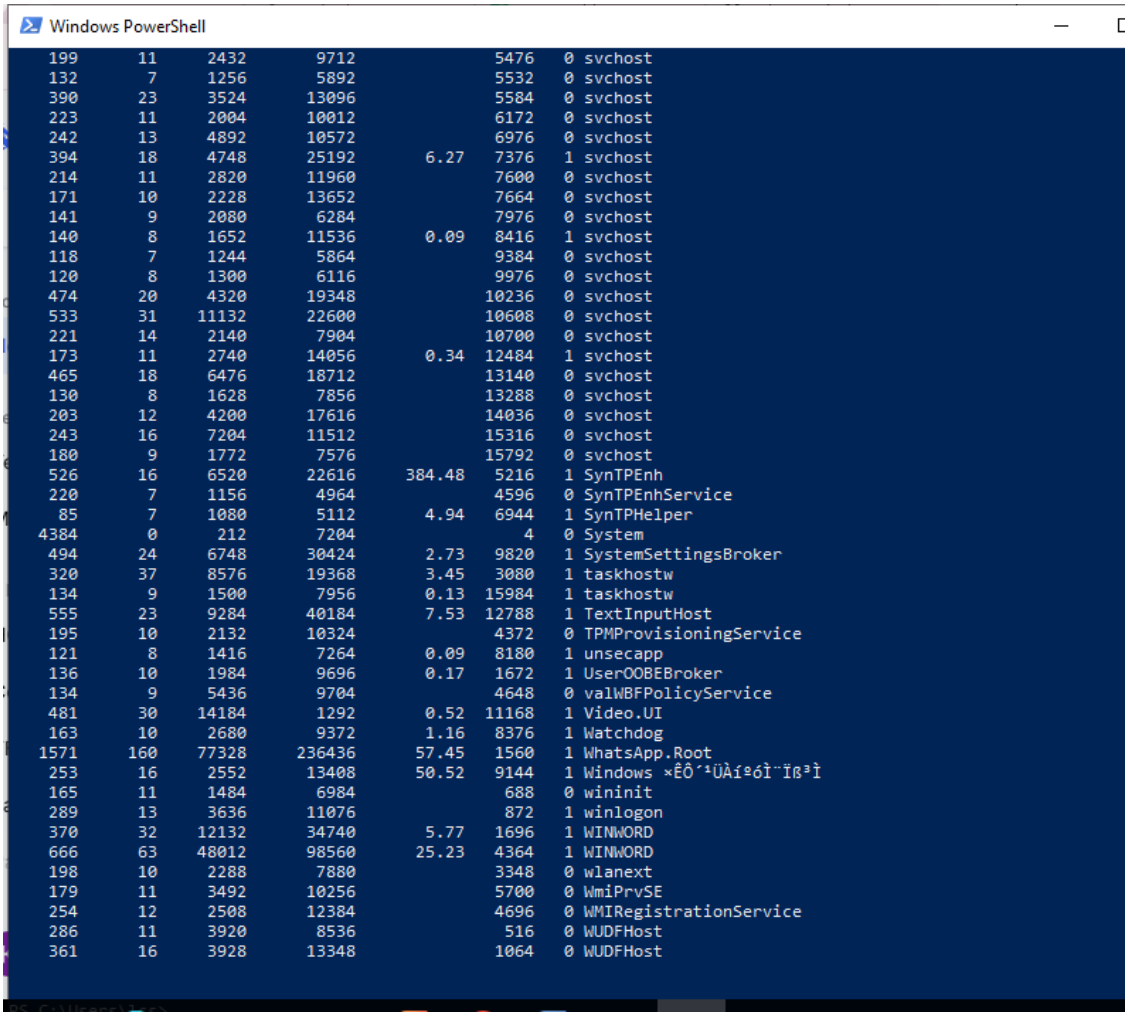



*Figure: Get-Process output*

**Explanation:** Shows running processes with handles, memory, CPU and process names.

## 5.7 Display system / computer information

```
Get-ComputerInfo
```

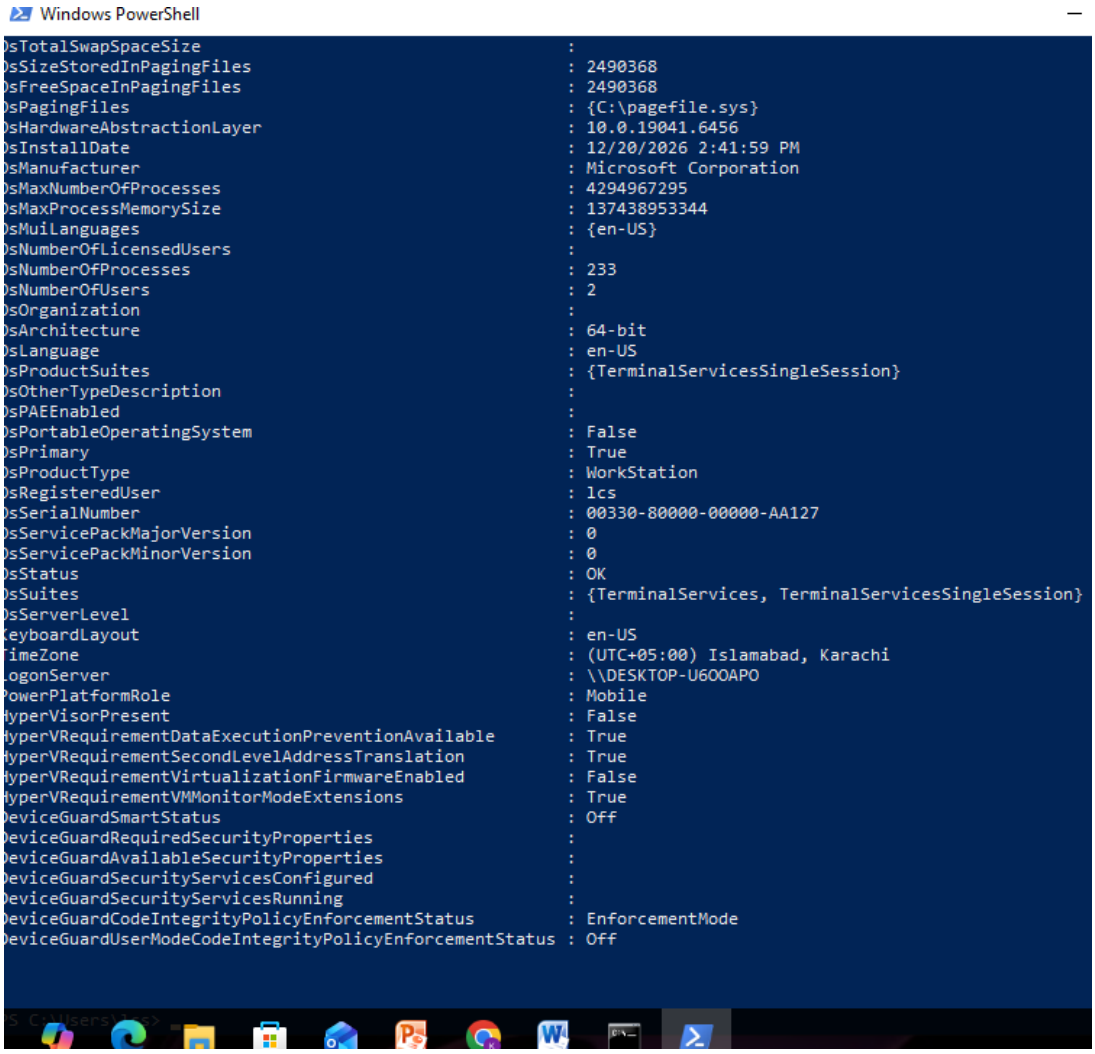



*Figure: Get-ComputerInfo output*

**Explanation:** Shows detailed OS and computer information such as Windows version, install date, manufacturer, time zone and RAM.

## 5.8 Copy the file into the Backup folder

```
Copy-Item "PowerShell_Task\info.txt" -Destination "Backup\info_backup.txt"
Get-ChildItem Backup
```

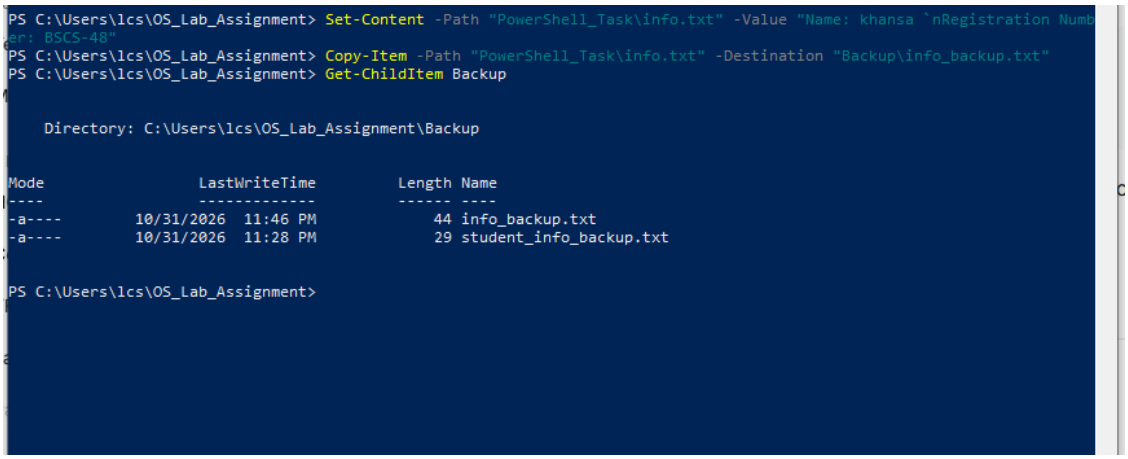



*Figure: Copying info.txt to Backup and listing Backup*

**Explanation:** Copy-Item copies the file; Get-ChildItem Backup proves info_backup.txt now exists in Backup next to student_info_backup.txt.

## 7. Task 6 — CMD vs PowerShell Comparison

CMD uses short, traditional commands such as dir, copy and ipconfig, while PowerShell uses a Verb-Noun style such as Get-Process and New-Item. CMD is simpler for quick tasks, but PowerShell is easier to learn in the long run because its names are consistent and readable. For automation, CMD depends on basic batch files with limited logic, whereas PowerShell scripts support variables, loops, functions and error handling. In system administration, CMD is good for simple file and network checks, but PowerShell works with objects, so output can be filtered, sorted and exported, and it can manage processes, services, the registry and remote computers. In my practical work, listing processes and getting system information was much more detailed with PowerShell. For advanced Windows administration I would prefer PowerShell because it is more powerful, scriptable and the modern Microsoft standard, while CMD remains useful for legacy commands and very simple batch files that need no extra setup.

(Word count: 154)

## 8. Conclusion

This assignment gave me practical experience of Windows administration. I learned that shortcuts such as Win + R, Win + X and Ctrl + Shift + Esc save a lot of time. With CMD I created a directory structure, managed files, and investigated the system through hostname, whoami, ipconfig, ping, tasklist and date/time commands. I automated routine work with the batch file system_check.bat using variables, conditions and file copying. Finally, I used PowerShell cmdlets for similar tasks and saw that PowerShell is more flexible and powerful. Overall, I gained confidence in using command-line tools and automation for operating-system administration.